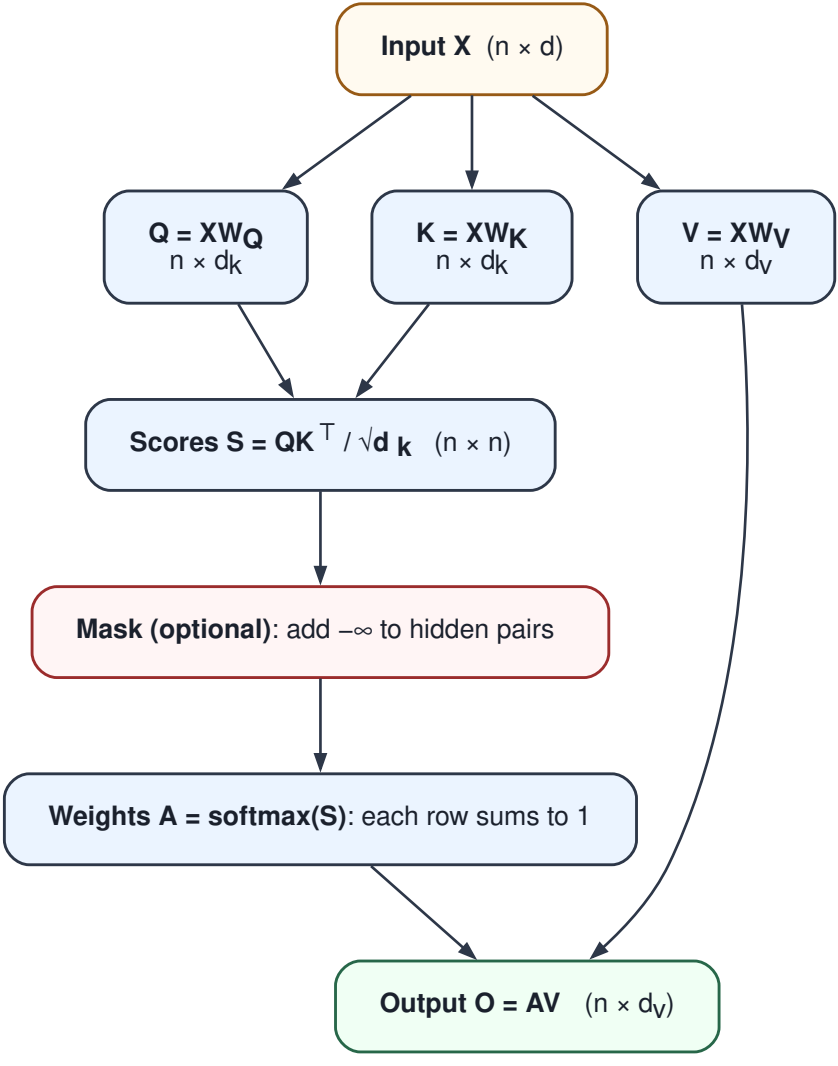
Input X (n × d)
Q = XW_Q
n × d_k
K = XW_K
n × d_k
V = XW_V
n × d_v
Scores S = QK^T / √d_k (n × n)
Mask (optional): add −∞ to hidden pairs
Weights A = softmax(S): each row sums to 1
Output O = AV (n × d_v)